

Learnify Platform Architecture Diagrams

C# / .NET Microservices E-Learning Platform

#	Diagram	Description
1	Entity-Relationship Diagram	MySQL schema — tables, PKs, FKs, constraints
2	UML Domain Class Diagram	Learnify.Core.Domain entities, enums & relationships
3	Service Architecture Diagram	Microservices topology, gateway routing & storage

1 — Entity-Relationship (ER) Diagram

Unified MySQL schema on Aiven Cloud — tables, primary keys, foreign keys, data types and constraints.

Identity Accounts		
PK Id	INT	
DisplayName	VARCHAR(100)	
EmailAddress	VARCHAR(255)	
PasswordHash	VARCHAR(500)	
Role	ENUM	

Identity Credentials		
PK Id	INT	
FK AccountId	INT	
Provider	VARCHAR(50)	
ExternalId	VARCHAR(255)	

Courses Catalog		
PK Id	INT	
Title	VARCHAR(200)	
Synopsis	TEXT	
Topic	VARCHAR(100)	
Difficulty	ENUM	
Language	VARCHAR(50)	

Curriculum Lessons		
PK Id	INT	
FK CourseId	INT	
Title	VARCHAR(200)	
Body	TEXT	
Format	ENUM	
MediaUrl	VARCHAR(500)	

Registration Enrollments		
PK Id	INT	
FK LearnerId	INT	
FK CourseId	INT	

Exams Quizzes		
PK Id	INT	
FK CourseId	INT	

Exams Attempts		
PK Id	INT	
FK QuizId	INT	
FK LearnerId	INT	
Score	INT	
IsPassed	BOOL	
AttemptedOn	DATETIME	

Tracking Progress		
PK Id	INT	
FK LearnerId	INT	
FK LessonId	INT	
WatchDurationMinutes	INT	
IsCompleted	BOOL	
LastAccessedOn	DATETIME	

Reviews Comments		
PK Id	INT	
FK CourseId	INT	
FK AuthorId	INT	
StarRating	INT(1-5)	
ReviewText	TEXT	
IsApproved	BOOL	
CreatedOn	DATETIME	

Analytics AuditLogs		
PK Id	INT	
Action	VARCHAR(100)	
EntityName	VARCHAR(100)	
EntityId	VARCHAR(100)	
BeforeValue	TEXT	
AfterValue	TEXT	
ActorId	INT	
Timestamp	DATETIME	

LEGEND	
PK	PK Primary Key
FK	FK Foreign Key

2 — UML Domain Class Diagram

Structural objects, enums, properties and navigation collections in Learnify.Core.Domain (EF Core).

LearnerAccount	
+ Id	int
+ DisplayName	string
+ EmailAddress	string
+ HashedPassword	string
+ Role	PlatformRole
+ IsActive	bool
+ RegisteredOn	DateTime
+ AuthoredCourses	ICollection<Course>
+ CourseRegistrations	ICollection<Reg>

CourseOffering	
+ Id	int
+ Title	string
+ Synopsis	string
+ AuthorId	int
+ Topic	string
+ Difficulty	DifficultyTier
+ ListPrice	decimal
+ IsPublished	bool
+ Author	LearnerAccount
+ Lessons	ICollection<Lesson>
+ Exams	ICollection<Exam>

CourseRegistration	
+ Id	int
+ LearnerId	int
+ CourseId	int
+ RegisteredOn	DateTime
+ Status	RegistrationStatus
+ CompletionPercent	int
+ CredentialIssued	bool
+ Learner	LearnerAccount
+ Course	CourseOffering

CourseLesson	
+ Id	int
+ CourseId	int
+ Title	string
+ Format	MediaFormat
+ MediaUrl	string
+ SequencePosition	int
+ DurationMinutes	int
+ IsPublished	bool
+ IsPreviewable	bool

LessonWatchRecord	
+ Id	int
+ LearnerId	int
+ LessonId	int
+ CourseId	int
+ IsFinished	bool
+ SecondsWatched	int
+ LastWatchedOn	DateTime

CourseExam	
+ Id	int
+ CourseId	int
+ Title	string
+ PassThreshold	int
+ AttemptsAllowed	int
+ IsPublished	bool
+ CreatedOn	DateTime

ExamAttempt	
+ Id	int
+ ExamId	int
+ LearnerId	int
+ BeganAt	DateTime
+ Score	int
+ HasPassed	bool
+ AnswersPayload	string

CourseFeedback	
+ Id	int
+ LearnerId	int
+ CourseId	int
+ StarRating	int
+ ReviewText	string
+ IsApproved	bool
+ SubmittedOn	DateTime

CredentialCertificate	
+ Id	int
+ CredentialCode	string
+ LearnerId	int
+ CourseId	int
+ AwardedOn	DateTime
+ DocumentUrl	string

AuditEntry	
+ Id	int
+ ActorId	int?
+ Action	string
+ EntityType	string
+ EntityId	string
+ OccurredAt	DateTime
+ IpAddress	string

AccountPlatformRole	
Learner	= 0
Instructor	= 1
Administrator	= 2

AccountDifficultyTier	
Beginner	= 0
Intermediate	= 1
Advanced	= 2
Expert	= 3

CourseRegistrationStatus	
Active	= 0

CourseMediaFormat	
Article	= 0

3 — Microservice Architecture Diagram

Full system topology: Client layer → YARP API Gateway → Microservices cluster → MySQL (Aiven) + Google OAuth.

